FOR

JOB RECRUITMENT System

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Appendix A: Glossary

Glossary

Term	Definition
SRS	Software Requirements Specification
HR	Human Resource
OTP	One-Time Password
CV	Curriculum Vitae
UI	User Interface
2FA	Two-Factor Authentication
API	Application Programming Interface
NFR	Non-Functional Requirement
FR	Functional Requirement

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Revision History

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Software Requirement Specification For Job Recruitment System

1. Introduction

1.1 Purpose

The purpose of the Job Requirement System is to **streamline the recruitment process** for organizations by allowing HR staff to post job openings, manage applications, and track candidates efficiently. The system will also enable job seekers to **search and apply for jobs online**, submit resumes, and receive notifications about application status. This system aims to reduce manual paperwork, save time, and improve the overall recruitment workflow for both employers and applicants.

1.2 Document Conversion

Term	Description
Shall	Refers to a mandatory requirement that must be fulfilled during Phase 1 of platform development
Should	Indicates a requirement that will take place in phase 2 and the developer should take it into consideration for future scalability
May	Refers to a requirement anticipated for Phase 3 or subsequent phases
Note	Provides additional information or clarification

Requirements are categorized as follows :-

Requirement Number	Description
FR-XX	Functional Requirements
NFR-XX	Non Functional Requirements

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders: -

Stakeholders	Role
Project Managers	Understand system scope
Developers / Programmers	Implement the system according to requirements
Testers / QA Team	Verify the system meets requirements
HR Staff / End Users	Use the system for recruitment operations

Reading Suggestions:

- Readers should focus on sections relevant to their role. For example, developers may focus on functional and non-functional requirements, while HR staff may focus on system features and user interactions.

- It is recommended to read the SRS in its entirety to gain a complete understanding of the system.

1.4 Product Scope

The Job **recruitment** System is designed to **automate and streamline the recruitment process** for organizations. Key features include:

- Posting and managing job openings by HR staff.
- Online application submission for candidates.
- Resume and document management.
- Search and filtering of candidates based on skills and experience.
- Notifications for application status updates.
- Reporting features for HR to track recruitment metrics.

Goals:

- Reduce manual paperwork and administrative effort.
- Improve candidate tracking and communication.
- Provide a user-friendly interface for both HR and applicants.

Out of Scope

The system will **not include** the following:

- Payroll management or employee onboarding beyond recruitment.
- Advanced AI-based candidate ranking or predictions.
- Mobile application (unless explicitly developed later).
- Background verification or reference checks.

1.5 REFERENCES

1.IEEE Std 830-1998, **IEEE Recommended Practice for Software Requirements Specifications**, IEEE Computer Society, 1998.

2.Pressman, R. S., & Maxim, B. R., **Software Engineering: A Practitioner's Approach**, 8th Edition, McGraw-Hill,

3.Sommerville, I., **Software Engineering**, 10th Edition, Pearson, 2015.

2. Overall Description

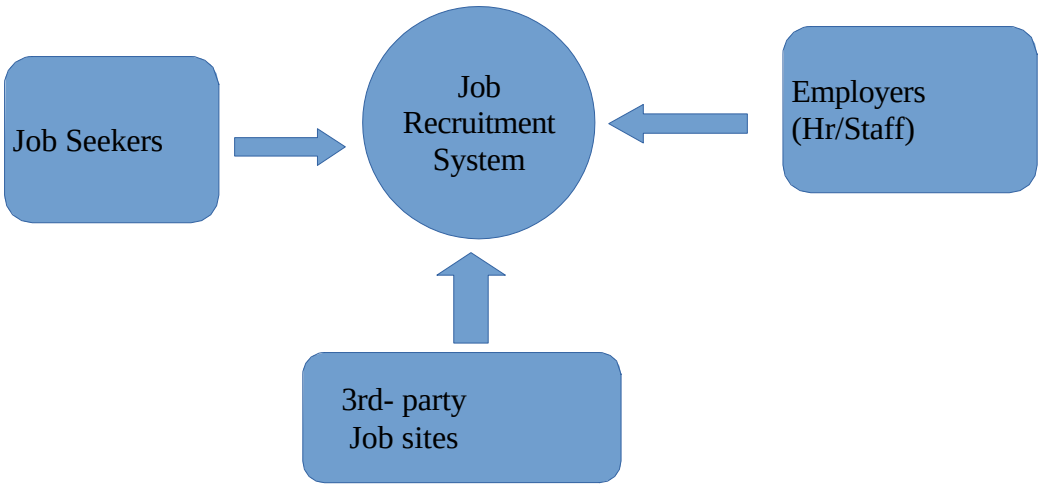
2.1 Product Perspective

The Job **recruitment** System is a **standalone web-based application** designed to support HR departments in managing recruitment processes. It is intended to operate independently but can be extended in the future to integrate with other HR systems, such as payroll or performance management systems.

Key Points:

- **Standalone System:** Does not require external software to function, aside from standard web browsers and a database server.
- **User Interface:** Provides separate interfaces for HR staff (admin portal) and job applicants (public portal).
- **Data Management:** Stores job postings, applications, resumes, and user information in a secure database.
- **Extensibility:** The system is designed to allow future integration with third-party services (e.g., LinkedIn, online testing platforms).

The following context diagram illustrates the system's position within its environment and its interactions with external entities:



2.2 Product Functions

The Job **recruitment** System provides the following key functions:

- **Job Posting Management:** HR staff can create, edit, and remove job vacancies, specifying job title, description, requirements, and deadlines.
- **Job Search and Filtering:** Candidates can search for jobs using keywords, location, skills, or department filters.
- **Online Application Submission:** Applicants can submit applications online and upload resumes or supporting documents.
- **Notifications:** The system sends automatic updates to applicants about the status of their applications.
- **Reporting and Analytics:** HR can generate reports on job postings, number of applicants, and recruitment metrics.
- **User Account Management:** Both HR staff and candidates can securely create and manage their accounts.
- **Security and Access Control:** Different permission levels ensure sensitive information is protected, with HR and candidate users having appropriate access.

2.3 User Classes, Characteristics, and Needs

The Job **recruitment** System will have the following user classes:

- **HR Staff / Administrators**
  - **Characteristics:** Familiar with recruitment processes, basic computer skills, responsible for managing job postings and candidate applications.
  - **Needs:** Easy-to-use interface to post and manage jobs, view applications, generate reports, and communicate with applicants efficiently.
- **Job Applicants / Candidates**
  - **Characteristics:** Individuals seeking employment, may have varying levels of computer literacy, need to submit applications and resumes online.
  - **Needs:** Clear navigation to search and filter job postings, simple application submission process, ability to track application status, and receive notifications.
- **System Administrators (IT Staff)**
  - **Characteristics:** Technical users responsible for system maintenance, security, and backups.

- **Needs:** Tools to manage users, maintain data integrity, perform security updates, and monitor system performance.

2.4 Operating Environment

The Job **recruitment** System is designed to operate in the following environment:

- **Hardware Requirements:**

  Server: Minimum 4-core CPU, 8 GB RAM, 500 GB storage.

  Client: Any desktop, laptop, or mobile device capable of running a modern web browser.

- **Software Requirements:**

  Server OS: Windows Server, Linux, or compatible.

  Web Server: Apache, Nginx, or similar.

  Database: MySQL, PostgreSQL, or any relational database.

  Frontend: Modern web browsers (Chrome, Firefox, Edge, Safari).

  Backend: Developed using a web framework (e.g., Django,

  FastAPI).

- **Network Requirements:**

  Internet or intranet access for users to interact with the system.

  Recommended minimum bandwidth: 1 Mbps for smooth operation.

- **Security Environment:**

  Secure HTTPS connection for data transfer.

  Role-based access control to protect sensitive information.

2.5 Design and Implementation Constraints

The Job **recruitment** System will be developed and deployed under the following constraints:

- **Platform Constraints:** The system must be fully web-based and accessible through major modern browsers (Chrome, Firefox, Edge, Safari) without requiring additional plugins or software installation. Mobile browser compatibility is recommended but not mandatory in the first phase.
- **Technology Constraints:** The backend will be developed using Django (Python) or FastAPI, and the database will be relational (MySQL or PostgreSQL). Frontend technologies will include HTML, CSS, and JavaScript. Any third-party libraries used must be compatible with the chosen framework.
- **Security Constraints:** All user data, including personal information and uploaded resumes, must be stored securely. Passwords must be hashed, and data transmission must use HTTPS. Role-based access control must be enforced to

separate HR and applicant permissions.

- **Performance Constraints:** The system should support at least 100 concurrent users without noticeable slowdowns. Page load time for users should not exceed 3 seconds under normal network conditions.
- **Regulatory and Compliance Constraints:** The system must comply with local and international data privacy laws, including secure handling of personal information for job applicants.
- **Hardware Constraints:** The server must have at least 4 CPU cores, 8 GB RAM, and 500 GB storage. The system should operate efficiently on standard office computers for HR staff and typical home devices for applicants.
- **User Interface Constraints:** The interface must be intuitive and simple for users with basic computer literacy. Forms, buttons, and notifications should follow usability best practices to minimize errors during job posting and application submission

2.6 User Documentation

The Job **recruitment** System will be accompanied by comprehensive user documentation to assist all types of users in effectively using the system. The documentation will include:

User Manual for HR Staff:

Step-by-step instructions on creating, editing, and deleting job postings.

Guidelines for reviewing, filtering, and managing candidate applications.

Instructions on generating recruitment reports and analytics.

Explanation of notification features and how to communicate with applicants.

User Manual for Job Applicants:

Instructions on creating and managing an account.

Guidelines for searching and filtering job

postings.

Step-by-step procedure for submitting applications and uploading resumes.

Information on tracking application status and receiving notifications.

Quick Reference Guide:

A condensed version of the manuals highlighting the mainfunctionalities and workflows for quick access.

Online Help / Tooltips:

Context-sensitive help within the system, including tooltips and pop-up instructions for forms, buttons, and other interactive elements.

Technical Support Information:

Contact details for helpdesk or technical support in case users encounter issues beyond the documentation.

2.7 Assumptions and Dependencies

The development and operation of the Job Requirement System are based on the following assumptions and dependencies:

- **Assumptions:**

  Users (HR staff and applicants) have basic computer literacy and access to a web browser.

  The organization has reliable internet connectivity for system access.

  HR staff will provide accurate job posting information and manage candidate responses appropriately.

  The system will be deployed in a secure server environment maintained by the organization or a trusted hosting provider.

  Users will follow proper authentication and security procedures, such as keeping passwords confidential.

- **Dependencies:**

  Web Browser Compatibility: The system depends on modern browsers (Chrome, Firefox, Edge, Safari) to function correctly.

  Database Server: The system requires a relational database (MySQL or PostgreSQL) to store job postings, applications, and user data.

  Web Framework: The system depends on the chosen backend framework (Django or FastAPI) for core functionalities.

  Email/Notification Service: The system relies on an email server or notification service to send status updates to applicants.

  Security Protocols: The system depends on HTTPS and proper server security configurations to protect sensitive user data.

3. External Interface Requirements

3.1 User Interfaces

The Job **recruitment** System will provide a user-friendly web-based interface accessible through modern browsers. The system will include the following UI components:

- **Login and Registration Pages:** Allow users to create accounts and authenticate securely.
- **HR Dashboard:** Displays job postings, applicant lists, notifications, and reporting tools.
- **Job Search Page:** Enables applicants to browse and filter job vacancies based on keywords, department, or location.
- **Application Submission Form:** Allows applicants to enter personal details and upload resumes or supporting documents.
- **Notification Panel:** Shows updates such as application status changes or new job posting alerts.

The interface will follow a clean, responsive layout that supports both desktop and mobile browsers.

3.2 Hardware Interfaces

The system does not require interaction with specialized external hardware. It relies on the following hardware environment:

- **Client Devices:** Standard desktop computers, laptops, or mobile devices capable of running a modern web browser.
- **Server Hardware:** A web server with minimum 4-core CPU, 8 GB RAM, and adequate storage for user data and uploaded documents.
  No additional hardware interfaces are required.

3.3 Software Interfaces

The system will interact with several software components to ensure proper functionality:

- **Operating Systems:** Compatible with Windows Server or Linux distributions.
- **Database Management System:** MySQL or PostgreSQL for storing user data, job postings, and applications.
- **Web Server Software:** Apache or Nginx to handle client requests.
- **Browsers:** Support for Chrome, Firefox, Edge, and Safari.

- **Email Service / SMTP Server:** Used for sending application status notifications to users. Any integration with external systems will follow standard API communication protocols if added in future versions.

3.4 Communications Interfaces

The system requires reliable communication channels for data exchange and user interaction:

- **Internet or Local Network Connection:** Required for accessing the web system.
- **Communication Protocols:** Uses HTTP/HTTPS for secure data transmission.
- **Email Protocols: Utilizes SMTP for sending automated emails to applicants and HR staff.**
- **Security Protocols:** SSL/TLS encryption will be used to protect sensitive data during communication.

4. System features and functional requirements

This section details the functional requirements of the Job **recruitment** system organized by major system features. Each requirement is identified with a unique ID (FR-XX) for traceability and refetence.

4.1 User Management and Authentication

4.1.1 Job Seeker Registration and Profile Management

ID	Requirement
FR-01	The system SHALL provide a self-registration process for job seekers.
FR-02	The system SHALL support account activation by user mobile number
FR-03	The system SHALL collect the following information from job seekers through a multi-level on boarding process (via mobile/email/desktop): Level 1: First and last name, email, mobile number, password, gender Level 2: Education, work experience, skills, training, certificates, recent salary (optional), current address, permanent address/residence Level 3 (optional): Social media links (e.g., LinkedIn,), personal statement, bio
FR-04	The system SHALL support resume/CV upload in common formats (PDF, DOCX, TXT).
FR-05	The system SHALL allow job seekers to build their profiles using a staged, stepwise form with popup prompts for the following sections: Education: Degree, institution Experience: Company, role, duration Skills: Primary/secondary; soft/hard Preferences: Job type, desired salary, preferred location
FR-06	The system SHALL allow job seekers to edit and update their profiles at any time.

FR-07	The system SHALL provide a recommendation to encourage job seekers to complete their profiles and to explore opportunities aligned with their broader skills, interests, and experiences beyond formal education and training.
FR-08	The system SHALL provide privacy controls that allow job seekers to set visibility preferences for their profile (public or private), and request account deactivation or deletion
FR-09	The system SHALL generate a shareable public profile URL/QR Code for each job seeker, if activated by the user.
FR-10	The system **SHOULD** allow job seekers to track who viewed their public profile and when (profile view analytics).
FR-11	The system **SHOULD** allow job seekers to import profile data automatically from external platforms such as LinkedIn or GitHub.
FR-12	The system **SHOULD** provide optional two-factor authentication (2FA) using email or mobile OTP to enhance account security.
FR-13	The system **SHOULD** allow job seekers to save multiple versions of their resumes/CVs for different job applications.

4.1.2 Employer Registration and Profile Management

ID	Requirement
FR-14	The system SHALL provide a stepwise registration process for employers: Level 1: Company name, email, mobile number, company ID, and registration number (if registered) Level 2: Website, industry, company size, address, company description, and attachments for supplementary documents
FR-15	The system SHALL require account activation through a one-time password (OTP) sent to the registered mobile number.
FR-16	The system SHALL display a Verified Employer badge next to the names of verified companies to enhance trust for job seekers and provide added value to employers who complete the verification process.
FR-17	The system SHALL provide a company profile page that showcases the employer's information, job openings, and company background
FR-18	The system SHALL allow employers to upload company logo, and images to enhance their profile. In addition to documents such as company registration and manage uploaded documents
FR-18	The system SHALL provide a dashboard for employers to manage job postings, view matched job seekers, create shortlists of interested or qualified candidates, and track key metrics.

4.1.3 Administrator User Management

ID	Requirements
FR-19	The system SHALL provide a comprehensive user management interface for administrators.
FR-20	The system SHALL allow authorized administrators to add and manage core entities on the platform, including job seekers, employers, and job offerings (opportunities)
FR-21	The system SHALL provide user management capabilities for administrators, including the ability to view, approve, ban, or deactivate user accounts, reset account credentials, and monitor account statuses.
FR-22	The system SHALL provide tools for administrators to assist users with account recovery and technical issues, view pending employer registrations and approve them based on submitted profile information.
FR-23	The system SHALL allow authorized administrators to define and update system settings if needed.

4.1.4 Authentication and Authorization

ID	Requirements
FR-24	The system SHALL implement secure authentication mechanisms including username/password, email verification, and multi-factor authentication options.
FR-25	The system SHALL enforce strong password policies with configurable parameters
FR-26	The system SHALL implement role-based access control to restrict access to features and data based on user roles.
FR-27	The system SHALL provide session management with configurable timeout settings.
FR-28	The system SHALL maintain detailed access logs for security monitoring and auditing.

4.2 Job Posting and Management

4.2.1 Job Creation and Publishing

ID	Requirement
FR-29	The system SHALL provide a structured job posting form for employers to create job listings.

FR-30	The system SHALL provide a structured job posting form that allows employers to add new job opportunities with the following fields: Job title, summary, and required skills (with support for file upload or copy/paste input) Contract type (e.g., full-time, part-time, training, project-based) Required education level Application deadline with an optional auto-close feature Work format selection (Physical with employment location, Online, or Hybrid) Gender, number of employees, etc
	Required languages and proficiency levels Link to the job if available. More detailed features on the job post to be provided.
FR-31	The system SHALL allow employers to manage their job postings by editing job details, extending application deadlines, and set job posting visibility (public, private, or targeted) posts when necessary

4.2.2 Job Search and Filtering

ID	Requirements
FR-33	The system SHALL provide a comprehensive search interface for job seekers to find relevant opportunities
FR-34	The system SHALL provide a search engine with advanced filtering capabilities, allowing job seekers to search for jobs using the following criteria: Keywords (text) Location Salary range Employment type Date posted Application deadline Job-specific requirements (e.g., skills, education, training, years of experience) Company sector/industry (e.g., IT, Health, NGO, Construction)
FR-35	The system SHALL also include a favorites feature (e.g., heart button) for users to save jobs of interest.
FR-36	The search interface SHALL support both basic (search by single keyword) and advanced search modes based on employer-defined attributes.

4.3 Job Application Management

ID	Requirement
FR-37	The system SHALL allow job seekers to apply for job postings directly through the platform.
FR-38	The system SHALL allow job seekers to submit applications using their uploaded CV, profile information, or multiple saved CV versions.
FR-39	The system SHALL prevent duplicate applications for the same job by the same job seeker.
FR-40	The system SHALL allow job seekers to view the real-time status of their applications (e.g., Submitted, Under Review, Shortlisted, Accepted, Rejected).
FR-41	The system SHALL notify job seekers via email/SMS when their application status changes.

FR-42	The system **SHALL send a notification (email/SMS/in-app)** to job seekers when their application is *Accepted*, *Shortlisted*, or *Rejected*.
FR-43	The system SHALL allow employers to review applications, download CVs, and update the application status.
FR-44	The system SHOULD support application screening or ranking based on skills, experience, or keyword matching (Phase 2 feature).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirements
NFR-01	The system SHALL provide page load times of less than 3 seconds for standard operations under normal load conditions.
NFR-02	The system SHALL provide search results within 2 seconds for standard search queries.
NFR-03	The system SHALL support at least 500 concurrent users during normal operations.
NFR-04	The system SHALL support at least 1,000 concurrent users during peak periods.
NFR-05	The system SHALL process at least 100 job applications per minute during peak periods.
NFR-06	The system SHALL support at least 500 new user registrations per day
NFR-08	The system SHALL optimize database queries to minimize I/O operations and response times.
NFR-09	The system SHALL implement caching mechanisms to reduce resource utilization for frequently accessed data.
NFR-10	The system SHALL be designed to scale horizontally by adding more server instances to handle increased load.

5.2. Security Requirements

ID	Requirements
NFR-11	The system SHALL enforce strong password policies, including minimum length, complexity, and regular password changes.
NFR-12	The system SHALL maintain detailed access logs for all authentication and authorization events.
NFR-13	The system SHALL automatically lock accounts after a specified number of failed login attempts.
NFR-14	The system SHALL implement secure session management with appropriate timeout settings.
NFR-15	The system SHALL encrypt all sensitive data at rest using industry-standard encryption algorithms
NFR-16	The system SHALL comply with Ethiopian data protection regulations

5.3 .Usability and Accessibility

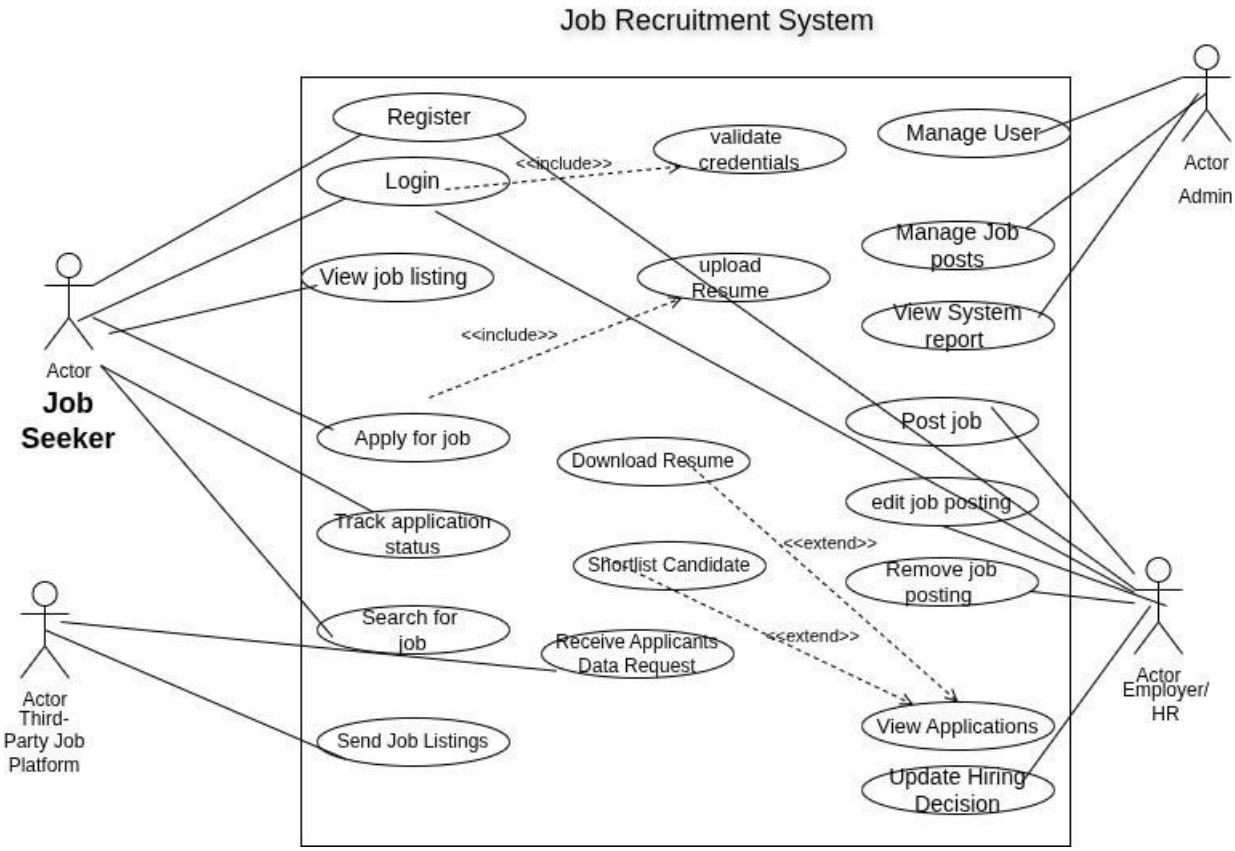
ID	Requirement
NFR-17	The system SHALL provide clear navigation and information architecture.
NFR-18	The system SHALL implement responsive design to support various screen sizes and devices.
NFR-20	The system SHALL provide clear navigation and information architecture.
NFR-21	The system SHALL ensure sufficient color contrast for text and interactive elements.
NFR-22	The system SHALL ensure that form elements have associated labels.
NFR-23	The system SHALL support screen readers and other assistive technologies.
NFR-24	The system SHALL minimize the number of steps required to complete common tasks.

5.4. Maintainability and Portability

NFR-25	The system SHALL be designed with a modular architecture to facilitate maintenance and updates
NFR-26	The system SHALL follow consistent coding standards and best practices.
NFR-27	The system SHALL include comprehensive technical documentation for all components.
NFR-29	The system SHALL be designed to operate in different hosting environments (on-premises, cloud, or hybrid)
NFR-30	The system SHALL minimize dependencies on specific hardware or operating system features
NFR-31	The system SHALL provide documented deployment procedures for different environments.
NFR-32	The system SHALL support automated deployment and configuration

5.5 .Reliability and Availability

ID	Requirements
NFR-33	The system SHALL implement high availability architecture to minimize single points of failure.
NFR-32	The system SHALL maintain 99.0% availability during non-standard hours
NFR-33	The system SHALL maintain system stability when encountering unexpected inputs or conditions.
NFR-34	The system SHALL provide advance notice of scheduled maintenance to all users.
NFR-35	The system **SHALL provide consistent performance** even when multiple users perform actions simultaneously.
NFR36	The system **SHALL automatically failover to backup services** if the primary server becomes unavailable.
NFR-37	he login and registration services **SHALL maintain high availability**, with downtime not exceeding **30 minutes per month**.

Appendix B: Analysis Models

Use Case Diagram